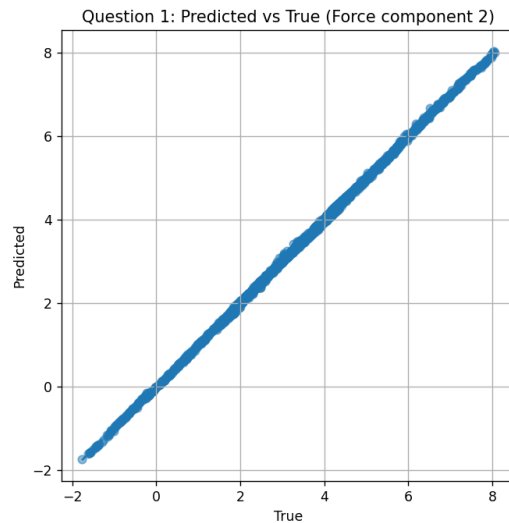
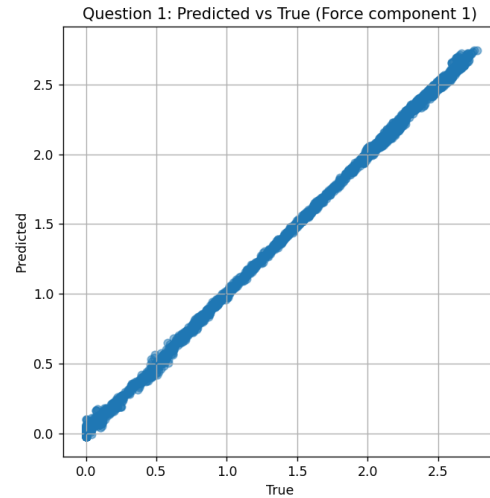
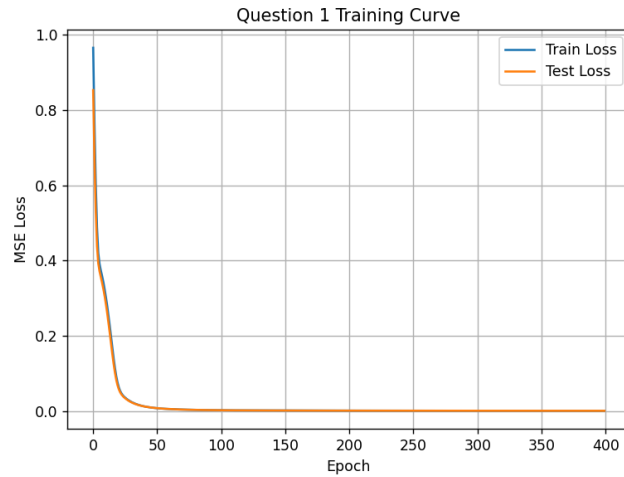


SYSM 6v80 Project 1 Discussion

Kayla Yhap

Question 1

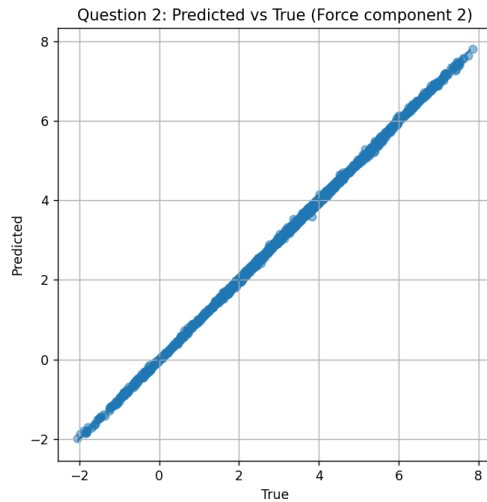
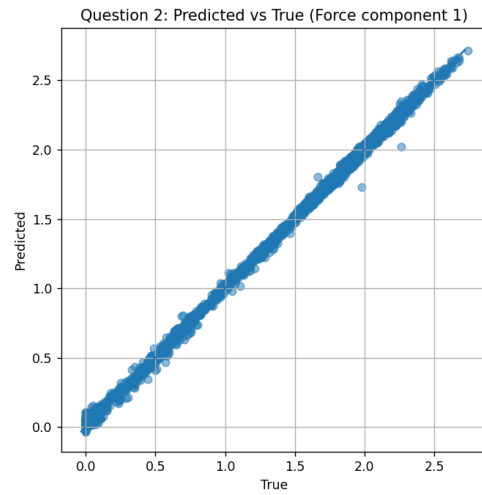
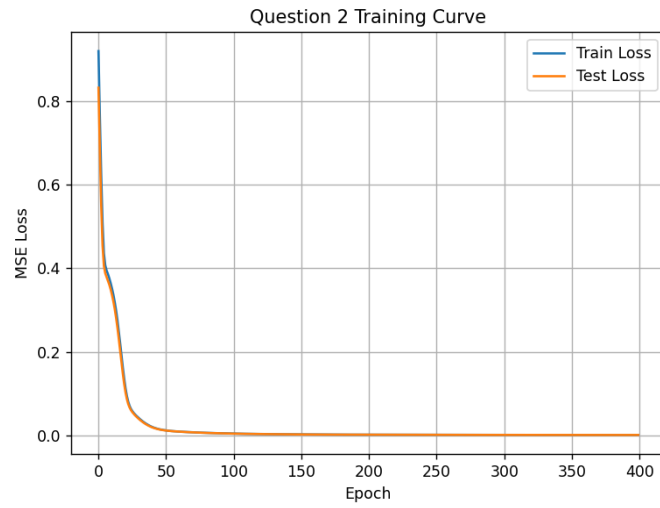


```
Epoch 0 | Train Loss = 0.965201 | Test Loss = 0.852638
Epoch 25 | Train Loss = 0.035431 | Test Loss = 0.033706
Epoch 50 | Train Loss = 0.007404 | Test Loss = 0.007474
Epoch 75 | Train Loss = 0.003233 | Test Loss = 0.003270
Epoch 100 | Train Loss = 0.001973 | Test Loss = 0.002006
Epoch 125 | Train Loss = 0.001495 | Test Loss = 0.001535
Epoch 150 | Train Loss = 0.001260 | Test Loss = 0.001280
Epoch 175 | Train Loss = 0.001083 | Test Loss = 0.001094
Epoch 200 | Train Loss = 0.000961 | Test Loss = 0.000977
Epoch 225 | Train Loss = 0.000836 | Test Loss = 0.000857
Epoch 250 | Train Loss = 0.000759 | Test Loss = 0.000799
Epoch 275 | Train Loss = 0.000698 | Test Loss = 0.000715
Epoch 300 | Train Loss = 0.000660 | Test Loss = 0.000669
Epoch 325 | Train Loss = 0.000627 | Test Loss = 0.000627
Epoch 350 | Train Loss = 0.000580 | Test Loss = 0.000601
Epoch 375 | Train Loss = 0.000550 | Test Loss = 0.000583
Epoch 399 | Train Loss = 0.000537 | Test Loss = 0.000548

Final metrics:
MAE: [0.01901456 0.02513984]
RMSE: [0.02391089 0.03330915]
```

The training and test loss are nearly identical and are very close to approaching 0. we can see this in question 1 training curve as well as in the predicted vs true forces where the relationship is near perfect and there is no overfitting. This is expected since our conditions are in static equilibrium. For all problems i have provided figures to qualitatively see the difference in loss when more complex conditions are applied.

Question 2:

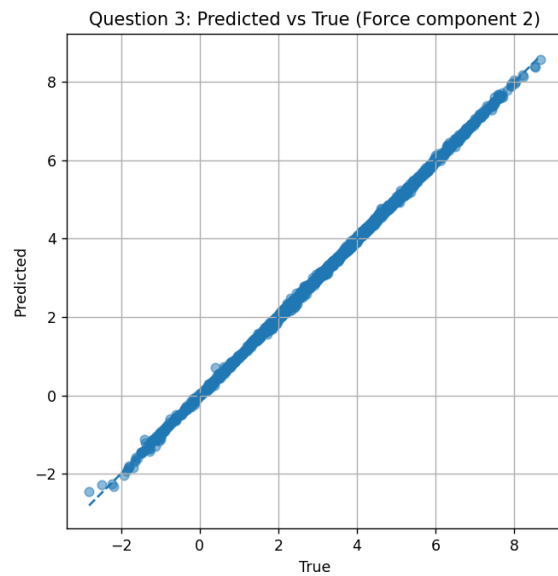
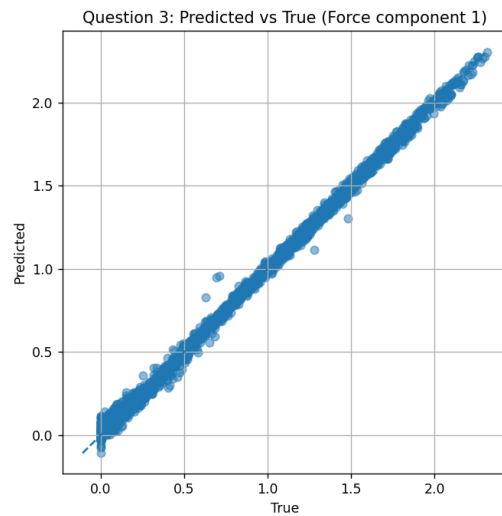
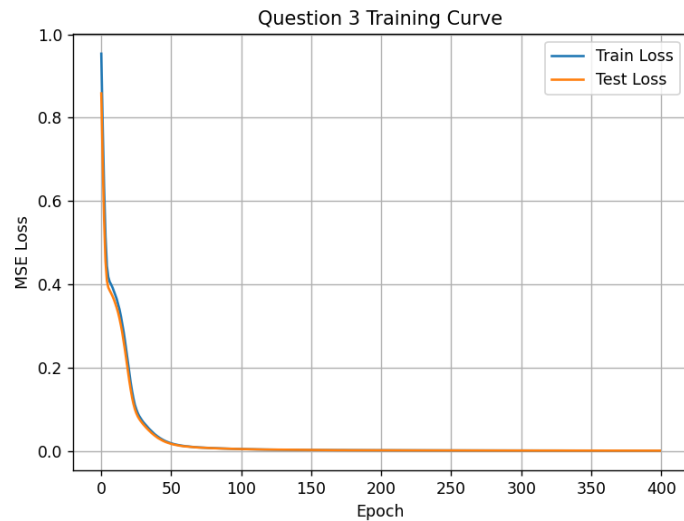


Epoch	0	Train Loss = 0.919769	Test Loss = 0.832169
Epoch	25	Train Loss = 0.058187	Test Loss = 0.055585
Epoch	50	Train Loss = 0.011797	Test Loss = 0.011502
Epoch	75	Train Loss = 0.006517	Test Loss = 0.006429
Epoch	100	Train Loss = 0.004201	Test Loss = 0.004138
Epoch	125	Train Loss = 0.002903	Test Loss = 0.002874
Epoch	150	Train Loss = 0.002174	Test Loss = 0.002186
Epoch	175	Train Loss = 0.001748	Test Loss = 0.001759
Epoch	200	Train Loss = 0.001473	Test Loss = 0.001494
Epoch	225	Train Loss = 0.001306	Test Loss = 0.001307
Epoch	250	Train Loss = 0.001172	Test Loss = 0.001198
Epoch	275	Train Loss = 0.001073	Test Loss = 0.001064
Epoch	300	Train Loss = 0.000997	Test Loss = 0.000999
Epoch	325	Train Loss = 0.000955	Test Loss = 0.000936
Epoch	350	Train Loss = 0.000919	Test Loss = 0.000912
Epoch	375	Train Loss = 0.000876	Test Loss = 0.000893
Epoch	399	Train Loss = 0.000861	Test Loss = 0.000894

Final metrics:
MAE: [0.02174507 0.02715539]
RMSE: [0.02870189 0.03629863]

While the training and test loss are near identical and approach 0, they are slightly higher than when the conditions were in static equilibrium. This is due to the mechanical system undergoing rotation at a constant angular velocity around a contact point. Even though the angular acceleration is 0 it introduces centripetal force that relies on thetadot^2 causing a more complex mapping and increasing the prediction error. We can see this in the predicted vs true force component figures where the points are more spread out.

Question 3:



```
Loading from: c:\Users\Kayla\Desktop\Project1SYSM6v80\data
Epoch 0 | Train Loss = 0.953538 | Test Loss = 0.858276
Epoch 25 | Train Loss = 0.100912 | Test Loss = 0.091518
Epoch 50 | Train Loss = 0.019006 | Test Loss = 0.017622
Epoch 75 | Train Loss = 0.008120 | Test Loss = 0.007960
Epoch 100 | Train Loss = 0.005254 | Test Loss = 0.005249
Epoch 125 | Train Loss = 0.003665 | Test Loss = 0.003743
Epoch 150 | Train Loss = 0.002854 | Test Loss = 0.002909
Epoch 175 | Train Loss = 0.002452 | Test Loss = 0.002495
Epoch 200 | Train Loss = 0.002250 | Test Loss = 0.002301
Epoch 225 | Train Loss = 0.002074 | Test Loss = 0.002098
Epoch 250 | Train Loss = 0.001914 | Test Loss = 0.001909
Epoch 275 | Train Loss = 0.001783 | Test Loss = 0.001732
Epoch 300 | Train Loss = 0.001679 | Test Loss = 0.001633
Epoch 325 | Train Loss = 0.001607 | Test Loss = 0.001581
Epoch 350 | Train Loss = 0.001496 | Test Loss = 0.001479
Epoch 375 | Train Loss = 0.001457 | Test Loss = 0.001403
Epoch 399 | Train Loss = 0.001398 | Test Loss = 0.001360

Final metrics:
MAE: [0.02287166 0.03338027]
RMSE: [0.03076741 0.04528281]
```

The training and test loss here is higher than the previous questions but still approaches 0. we can also see in the training curve, they are not as identical as they were in the previous questions. This is expected though because the contact point has to move with a constant velocity which would cause nonlinear behavior. As we can see in the figure, the predictions are more spread out, showing that it was harder for the model to learn yet there is no overfitting so it still is learning well.

Comparison of Q1-Q3:

The results show that as the systems become more complicated, it is harder for the model to learn. We see this in Q1 where the system is at static equilibrium, it is easy for the model to learn whereas in question 2, the system becomes dependent on the velocity causing a more nonlinear system. In Q3 we added a constraint to the contact point which will increase the nonlinearity and cause higher loss. All of this can be seen in the figures where the points become more spread out the more complex the system gets

Question 4:

$$1) \quad A = \begin{bmatrix} -1 & -s_\theta & 0 & 1 \\ 0 & c_\theta & 1 & 0 \\ (l - \frac{h}{c_\theta}) & l(s_\theta - c_\theta) & 0 & -l \end{bmatrix} \quad \begin{matrix} \ddot{\theta} = 0 \\ \alpha = \dot{\theta} = 0 \end{matrix}$$

$$f_g + \sum f_i = m a = 0$$

$$f_g = -mgz_w = -mg(-s_\theta x_b + c_\theta z_b)$$

$$f_1 = \varphi_1 x_b$$

$$f_2 = \varphi_2 z_w = -\varphi_2 s_\theta x_b + \varphi_2 c_\theta z_b$$

$$f_3 = \varphi_{3,x} x_b + \varphi_{3,z} z_b$$

$$0 = mg s_\theta x_b - mg c_\theta z_b + \varphi_1 x_b - \varphi_2 s_\theta x_b + \varphi_2 c_\theta z_b + \varphi_{3,x} x_b + \varphi_{3,z} z_b$$

$$x_b: 0 = mg s_\theta + \varphi_1 - \varphi_2 s_\theta + \varphi_{3,x}$$

$$-mg s_\theta = \varphi_1 - \varphi_2 s_\theta + \varphi_{3,x}$$

$$z_b: 0 = -mg c_\theta + \varphi_2 c_\theta + \varphi_{3,z}$$

$$mg c_\theta = \varphi_2 c_\theta + \varphi_{3,z}$$

$$I\alpha = \sum p_i \times f_i$$

$$0 = p_1 \cdot f_1 + p_2 \cdot f_2 + p_3 \cdot f_3$$

$$\left(l - \frac{h}{c_\theta}\right) \varphi_1 + l(s_\theta - c_\theta) \varphi_2 + 0 \varphi_{3,z} - l \varphi_{3,x} = 0 = I\alpha$$

$$F_1 = \begin{bmatrix} -mg s_\theta \\ mg c_\theta \\ 0 \end{bmatrix}$$

This system is not in motion, so there are no additional forces meaning that this system is an equilibrium and is only affected by gravity

$$2) \quad a = \left[\left(l - \frac{2h}{l_0} \right) \ddot{\theta} + \left(l - \frac{2ht_0}{l_0} \right) \dot{\theta}^2 \right] x_b + \left[l \ddot{\theta} - \left(l - \frac{2h}{l_0} \right) \dot{\theta}^2 \right] z_b$$

$$\alpha = \ddot{\theta} = 0$$

$$\dot{\theta} = \bar{\omega}$$

$$a_{xb}: 0 + \left(l - \frac{2ht_0}{l_0} \right) \bar{\omega}^2$$

$$F = m \left[\left(l - \frac{2ht_0}{l_0} \right) \bar{\omega} \right] - mg \sin \theta$$

$$a_{zb}: 0 - \left(l - \frac{2h}{l_0} \right) \bar{\omega}^2$$

$$F = m \left[\left(-l + \frac{2h}{l_0} \right) \bar{\omega}^2 \right] + mg \cos \theta$$

$$I \alpha = 0$$

$$F_2 = \begin{bmatrix} m \left[\left(l - \frac{2ht_0}{l_0} \right) \bar{\omega} \right] - mg \sin \theta \\ m \left[\left(-l + \frac{2h}{l_0} \right) \bar{\omega}^2 \right] + mg \cos \theta \end{bmatrix}$$

In this system, the angular acceleration is zero but the box is rotating around a fixed contact point so there is still centripetal forces, which is why we have $\dot{\theta}^2$ squared

$$3) \quad a = \left[\left(l - \frac{2h}{C_\theta} \right) \ddot{\theta} + \left(l - \frac{4ht_\theta}{C_\theta} \right) \dot{\theta}^2 \right] x_b + \left[(l t_\theta) \ddot{\theta} - \left(l + \frac{4ht_\theta^2}{C_\theta} \right) \dot{\theta}^2 \right] z_b$$

$$\ddot{\theta} = 0$$

$$\dot{\theta} = \bar{\omega}$$

$$a_{xb}: 0 + \left(l - \frac{4ht_\theta}{C_\theta} \right) \bar{\omega}^2$$

$$f = m \left(l - \frac{4ht_\theta}{C_\theta} \right) \bar{\omega}^2 - mg \delta_\theta$$

$$a_{zb}: 0 - \left(l + \frac{4ht_\theta^2}{C_\theta} \right) \bar{\omega}^2$$

$$f = m \left(-l - \frac{4ht_\theta^2}{C_\theta} \right) \bar{\omega}^2 + mg C_\theta$$

$$\Sigma \alpha = 0$$

$$F_z = \begin{bmatrix} m \left(l - \frac{4ht_\theta}{C_\theta} \right) \bar{\omega}^2 - mg \delta_\theta \\ m \left(-l - \frac{4ht_\theta^2}{C_\theta} \right) \bar{\omega}^2 + mg C_\theta \end{bmatrix}$$

The system is now rotating and moving at the same time while keeping the contact point constrained, making the force heavily depend on the dot squared and it makes the geometry and constraint of the contact point hard to control